

[Insert Your Title Here]

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Abstract—[Insert Abstract Here: Provide a brief summary of the project, methods, and key results.]

I. INTRODUCTION

[Insert the introduction here. Provide background information and the motivation for the research.]

II. OBJECTIVES

The key objectives of this project are:

- 1) **Objective 1:** [Insert Objective 1 Here]
- 2) **Objective 2:** [Insert Objective 2 Here]
- 3) **Objective 3:** [Insert Objective 3 Here]

III. EXPLORATORY DATA ANALYSIS (EDA)

[Provide details on how the dataset was explored and analyzed before implementing the classifier.]

A. Data Distribution

[Discuss the distribution of the data points for each class in the feature space.]

B. Cluster Formation

[Insert description of how clusters were formed based on the data distribution.]

IV. DATA PROCESSING

[Outline the steps of preprocessing the data.]

A. Normalization/Standardization

[Insert the approach used for normalizing or standardizing the data.]

B. Data Splitting

[Describe how the data was split into training and testing sets.]

V. BAYESIAN CLASSIFIER

A. Multivariate Gaussian Distribution

[Provide the formulation for the multivariate Gaussian distribution.]

$$f(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

B. Posterior Calculation

[Explain the process of posterior calculation using Bayes' theorem.]

VI. RESULTS AND OBSERVATIONS

- **Precision per class:** [Insert Precision Results]
- **Recall per class:** [Insert Recall Results]
- **Mean Recall:** [Insert Mean Recall]
- **F-Measure per class:** [Insert F-Measure Results]
- **Classifier Accuracy:** [Insert Accuracy]

VII. CONCLUSION

[Summarize the findings, conclusions, and possible future work.]

ACKNOWLEDGMENT

[Insert acknowledgment text. Mention any individuals or institutions that contributed to the project.]

REFERENCES

- [1] [Insert citation 1 here]
- [2] [Insert citation 2 here]
- [3] [Insert citation 3 here]
- [4] [Insert citation 4 here]
- [5] [Insert citation 5 here]